

Classroom activities

Questions

Q1. Define Autoencoder, Draw the architecture of Autoencoder and execute autoencoder for a simple application of your own.

Q2. What is a Variable Autoencoder (VAE) with a simple example of your own.

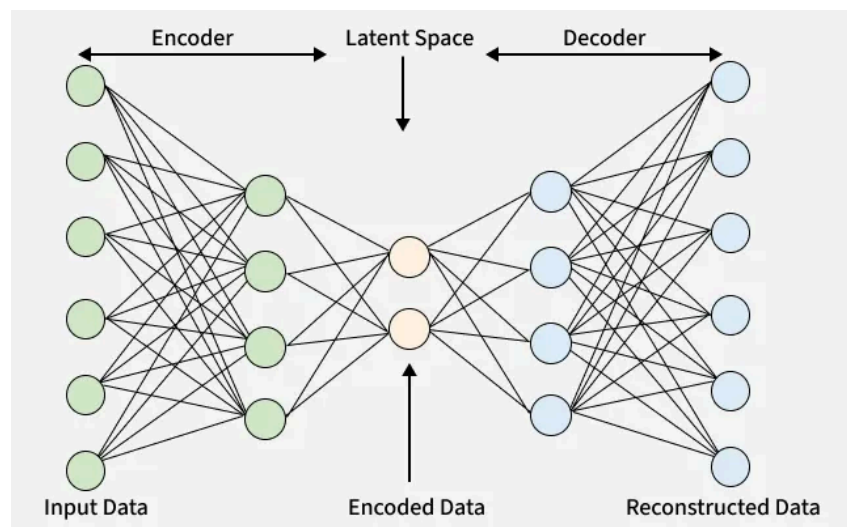
Q3. Compare Autoencoder vs variable Autoencoder.

Answers

Q1. An Autoencoder is an unsupervised neural network used to learn efficient data representations (encoding) by compressing input data into a lower dimensional form and reconstructing it back to the original input.

Key Components and Architecture

- Encoder: Compresses the input data into a smaller, compact representation (latent space/bottleneck layer).
- Bottleneck: The layer containing the compressed representation of the input data
- Decoder: Reconstructs the input data from the compressed latent space representation.
- Reconstruction Loss: The objective function that measures how closely the output matches the input, which the model aims to minimize.



Basic python implementation

```
import tensorflow as tf
from tensorflow.keras import layers, models
import numpy as np
from tensorflow.keras.datasets import mnist

# Load dataset
(x_train, _), (x_test, _) = mnist.load_data()
x_train = x_train / 255.0
x_test = x_test / 255.0

# Add noise
noise_factor = 0.5
x_train_noisy = x_train + noise_factor *
np.random.normal(size=x_train.shape)
x_test_noisy = x_test + noise_factor * np.random.normal(size=x_test.shape)

# Autoencoder Model
input_img = layers.Input(shape=(28,28))

# Encoder
x = layers.Flatten()(input_img)
encoded = layers.Dense(64, activation='relu')(x)

# Decoder
x = layers.Dense(784, activation='sigmoid')(encoded)
decoded = layers.Reshape((28,28))(x)

autoencoder = models.Model(input_img, decoded)

autoencoder.compile(optimizer='adam',
                    loss='binary_crossentropy')

# Training
autoencoder.fit(x_train_noisy, x_train,
                epochs=5,
                batch_size=256,
                validation_data=(x_test_noisy, x_test))
```

Q2. A Variational Autoencoder (VAE) is a generative deep learning model that compresses input data into a probabilistic, continuous latent space, allowing it to both reconstruct data and generate new, similar samples.

Key Components of Variational Autoencoder (VAE)

- **Encoder Network**
 - Converts input data into probability distribution.
 - Outputs mean (μ) and variance (σ) instead of fixed values.
- **Latent Space**
 - Compressed probabilistic representation of input data.
 - Usually follows a normal (Gaussian) distribution.
- **Sampling Layer (Reparameterization Trick)**
 - Samples latent vector using:
$$z = \mu + \sigma \times \epsilon \quad z = \mu + \sigma \times \epsilon$$
 - Enables backpropagation during training.
- **Decoder Network**
 - Reconstructs input data from sampled latent vectors.
 - Can also generate new data samples.
- **Loss Function**
 - **Reconstruction Loss:** Measures similarity between input and output.
 - **KL Divergence Loss:** Forces latent space to follow normal distribution.
- **Overall Objective**
 - Learn data distribution and generate realistic new samples.

Application: Generate new handwritten digits.

Example:

- Train VAE using digit images.
- The model learns digit distribution.
- Sampling random latent vectors generates new digits never seen before.

Q3.

Feature	Autoencoder (AE)	Variational Autoencoder (VAE)
Definition	Neural network used to learn compressed representation of data	Probabilistic autoencoder that learns data distribution
Latent Space	Fixed encoded vector	Learns probability distribution (mean & variance)
Output Nature	Deterministic output	Stochastic (random sampling involved)
Encoding	Maps input → latent vector	Maps input → mean (μ) and variance (σ^2)
Sampling	No sampling	Uses sampling (reparameterization trick)
Goal	Data reconstruction	Data generation + reconstruction
Loss Function	Reconstruction loss only	Reconstruction loss + KL Divergence loss
Generative Ability	Cannot generate new realistic data well	Can generate new data samples
Applications	Compression, denoising, feature learning	Image generation, anomaly detection, data synthesis
Latent Space Structure	Unstructured	Continuous & smooth structured space